



AGRICULTURE

6882/02

Paper 2

Specimen Paper October/November 2027-2029

Confidential

MARK SCHEME

SECTION A: Structured Questions (70 marks)

- 1 (a) improve crop and livestock production techniques; develop new machinery, tools, equipment; improved varieties; improve pastures and fodder quality; develop techniques for pest and disease control; improved inputs; informs policy making **(any two)** [2]
- (b) (i) growing crops in a nutrient solution [1]
- (ii) produce is clean; eliminates soil borne diseases; no nutrient losses; conserves water; suitable in areas with limited land/ poor soils **(any one point) R = high yields** [1]
- (c) enhance food security/more produce/yields; more exports/ ensuring foreign exchange; creates employment/ creates jobs **(any two)** [2]
- (d) depletion of nutrients; same nutrient demand yearly [2]
- (e) costs of certification of inputs and registration; high labour cost/ labour intensive; cost of rehabilitation of land **(any two) R= cost of organic inputs** [2]
- [10 marks]**
2. (a) the art and science of organising limited resources/ use of minimum inputs to achieve maximum returns (commodities produced by farming) [1]
- (b) fund seasonal agricultural operations such as cultivation/ pest control; purchase inputs/land; commercialise farming/ expanding scale o production **(any two)** [2]
- (c) purpose of loan; business plan; tax clearance; business experience; credit history; personal information/company profile; audited financial statements; collateral; information on outstanding loans **(any two)** [2]
- (d) are assets that are essential in the production process; includes tools, machinery and equipment, farm inputs and money [2]
- (e) decision making tool; comparing enterprises; planning; sourcing of inputs; accessing a loan. **(any two)** [2]

[9 marks]

3. (a) (i) intercropping **A = mixed cropping** [1]
- (ii) prevent soil erosion; beans act as cover crop; nitrogen fixation; beans act as living
Mulch; suppress weeds; conserves moisture **(any two)** [2]
- (b) poor aeration; poor root respiration; poor microbial activity; poor infiltration/ poor drainage;
reduced pore spaces; particles tightly packed; poor root development [2]
- (c) temperature; light intensity; humidity; carbon dioxide; wind; rainfall **(any two)** [2]
- (d) register with Eswatini Revenue Services; export declaration form; exchange control
form; invoice, road manifest; export permits/ trading license/ market rights; certificate of
origin; certificate of inspection of goods/ food safety adherence, relevant packaging;
transport unit sealed by Customs/quality control maintenance; phytosanitary certificates/
disease/ pest free products **(any two)** [2]
- [9 marks]**
4. (a) (i) 30kg [1]
- (ii) 135kg [1]
- (iii) variety A is 2.0 units - variety B is 1.85 units = 0.15 unit [2]
- (b) random sampling [1]
- (c) soil pollution/ chemical residues in soils; soil acidity/ some chemical fertilisers decrease pH;
pollution of water bodies/ eutrophication; destroy soil structure; reduced microbial activity [2]
- (d) over irrigation/too much watering; leaching; application of inorganic fertilisers/ ammonium
sulphate; application of poultry litter; removal of crop residues; acid rain [2]
- (e) less H ions promote microbial activity; less pH promote nutrients availability/ change solubility [2]
- [11 marks]**
5. (a)(i) seed coat/ testa [1]
- (ii) epigeal germination [1]
- (b) suitable temperature/ warmth; oxygen; moisture; viable seed **(any two)** [2]
- (c) water evaporates from the leaves' surfaces; a negative pressure is created; water is
pulled up from the roots to the upper part of the plant including the leaves/transpiration pull. [3]
- (d) new plants identical to parent; cheap/ planting material readily available;
mature early; hardy **(any three)** [3]
- [10 marks]**

6. (a) proboscis/ stylet [1]

(b) (i) reduces plant sap; causes plant to wilt and die; nutrient distribution affected;
reduces surface area for leaf to photosynthesis. (any two) [2]

(ii) absorbed into sap/ translocated in the phloem; kills pests feeding on poisoned sap [2]

(c) (i) 1.6 kg = 1 ha
X = 0.45 ha

$$x = \frac{1.6 \text{ kg} \times 0.45 \text{ ha}}{1 \text{ ha}}$$

$$x = 0.72 \text{ kg}$$

(1 mark working; 1 mark answer)

[2]

(ii) 1.6 kg = 200 litres

$$x = 8 \text{ litres}$$

$$x = \frac{1.6 \text{ kg} \times 8 \text{ litres}}{200 \text{ litres}}$$

$$x = 0.064 \text{ kg or } 64 \text{ grams}$$

(1 mark working; 1 mark answer)

[2]

(d) methods of pest control that do not harm the environment, which starts on diagnosis of the nature and source of pest problems;

preventative tactics include use predation/parasitism/ herbivory/other natural mechanisms; importation/ augmentation/ conservation of natural enemies **OR**

proactive tactics include use crop rotation/ resistant crop cultivars/ disease-free seed/ plants/ crop sanitation/ spacing of plants/ altering planting dates/ mulches (1 mark definition; 1 mark example) [2]

[11 marks]

7. (a) contains all nutrients required for healthy growth/ has all required nutrients in correct proportions [1]

(b) replacing worn out tissues; growth; reproduction; production (any two) [2]

(c) lubricant; solvent; regulates body temperature; aids digestion (any two) [2]

(d) (i) B – pancreas [1]

(ii) most digestion occur; absorption of nutrients (any two) [1]

(iii) regurgitation; microorganisms; fermentation; churning and mixing [3]

[10 marks]

SECTION B - Essay (30 marks)

8. (a) inbreeding; mating closely rabbits of the same breed;
crossbreeding; mating two different breeds
outcrossing; mating unrelated rabbits of the same breed **(any two)** [4]
- (b) before and after mating; 120g/day for maintenance
10 days before kindling; 125g/day to nourish the foetus/ body reserves
After kindling; as much as they desire until weaning from lactation [6]
- (c) adapted to local climatic conditions; higher temperatures; disease resistance; hardy/ survives
harsh conditions; high fertility; suits local farmers' needs [5]
- [15 marks]**

9. (a) roof; into water tanks
catchment area; into dams
rivers/ streams; dams **any two** [4]
- (b) mulching reduces evaporation; minimum tillage reduces soil exposure; shading reducing direct sunlight; hedge/live mulch reduces evapotranspiration **(any three points; with explanations)** [6]
- (c) increases photosynthesis; increase transpiration so speeds up growth; increase enzyme activity/metabolism; hastens growth stages; early maturity [5]
- [15 marks]**

10. (a) building faces away from direct sunlight; build against direction of prevailing wind/ blind wall should face oncoming wind; distance away from living quarters to reduce smell/ noise pollution; near water and power supply; accessibility for inputs delivery and marketing; gentle sloping land to avoid flooding; use poor soils as fertile soil used for crops; avoid site obstructions; adhere to local government regulations for security and political stability
(any two points explained) [4]
- (b) **soaking end method**; posts put into oil drum containing creosote for 3 days; more effective when the creosote is heated to almost boiling point and allowed to cool; heat improves sucking in of creosote into posts; turn posts upside down and leave for another 3 days; allow posts to drain in empty drum; allow posts to dry to prevent termite infestation

OR

Sap displacement method/ end diffusion method; dip freshly cut posts into a drum full of creosote; allow fresh wood to suck creosote into inner tissues; switch sides for wood to allow both ends to suck creosote; creosote will then displace the sap in the cells of the tissues; allow wood to dry

OR

Pressure method/ chamber method; metallic cylinder provides; pressure exerted on the

posts forces creosote into the wood; release pressure to allow sap to drain out; creosote replaces sap to treat the wood; allow wood to drain

[6]

(c) cheap; easy to plant; serves as windbreaks; decorative; controls soil erosion

[5]

[15 marks]